Best model evaluation

this is my result of the models table

based on this which is the best and why

Model	MAE	MSE	RMSE	R^2	MAPE (%)
Linear Regression	9.4	175.70	13.25	0.191040	9.17
Random Forest	9.08	165.73	12.87	0.241600	8.83
XGBoost	9.78	185.26	13.61	0.146234	9.61
Artificial Neural Network (ANN)	9.45	168.82	12.99	0.213652	9.19
BiLSTM	10.21	198.62	14.09	0.107857	10.03
K-Nearest Neighbors (KNN) Regressor	9.73	181.11	13.45	0.175010	9.52
Support Vector Machine Regression (SVR)	9.82	194.18	13.93	0.137424	9.49
AdaBoost Regression	9.84	170.29	13.04	0.230850	9.87

Looking at your table, we can evaluate the models based on the standard regression metrics you provided: **MAE, MSE, RMSE, R^2, and MAPE**. Let's break it down carefully:

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1. Error Metrics (Lower is Better)

- **MAE (Mean Absolute Error)**: Measures average absolute difference between predicted and actual values.
  - **Lowest:** Random Forest (9.08) → best
  - ANN (9.45) and Linear Regression (9.4) are close behind.
- **MSE / RMSE**: Penalizes larger errors more heavily.
  - **Lowest RMSE:** Random Forest (12.87) → best
  - ANN (12.99) and AdaBoost (13.04) follow.
- **MAPE (%):** Percentage error.
  - **Lowest:** Random Forest (8.83%) → best

✅ Conclusion from error metrics: **Random Forest consistently has the lowest errors**, so it predicts more accurately on average.

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2. R^2 (Coefficient of Determination) – Higher is Better

- Shows how much variance in the target variable is explained by the model.
  - **Highest R^2:** Random Forest (0.2416) → best
  - AdaBoost (0.23085) and ANN (0.21365) are next.

Even the best R^2 is low (~0.24), meaning the model explains ~24% of variance—your dataset might be noisy or limited, but comparatively RF still performs better.

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3. Other Observations

- BiLSTM and SVR are underperforming: higher errors and low R^2.
- Linear Regression is simple and interpretable but slightly worse than Random Forest in accuracy.
- Ensemble methods (Random Forest, AdaBoost) generally perform better because they reduce overfitting and combine multiple learners.

✅ Overall Recommendation

Random Forest is the best model here because:

1. **Lowest MAE, RMSE, and MAPE** → most accurate predictions.
2. **Highest R^2** → explains more variance than other models.
3. Ensemble method → more robust than simple linear or single neural network models.

If you want, I can also suggest **how to further improve Random Forest performance** on your dataset. Do you want me to do that?

teach me about all the metrics and how we are understading the model performnce.

Absolutely! Let's go **step by step** and really understand each metric and what it tells us about model performance. I'll explain it in a way that connects directly to your table.

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1. MAE – Mean Absolute Error

Formula:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

- y_i = actual value
- $\hat{y}_i$ = predicted value
- n = number of samples

Meaning:

- It's the **average absolute difference** between predicted and actual values.
- Every error counts equally.

- Lower MAE → better model.

Example from your table:

- Random Forest MAE = 9.08 → on average, its predictions are **~9 units away from the true value**.
- BiLSTM MAE = 10.21 → less accurate.

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2. MSE – Mean Squared Error

Formula:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Meaning:

- Squares the errors → **punishes larger errors more heavily**.
- Good for situations where big mistakes are much worse than small mistakes.
- Lower MSE → better.

Example:

- Random Forest MSE = 165.73 → smaller than others.
- BiLSTM MSE = 198.62 → big errors occur more often.

---

3. RMSE – Root Mean Squared Error

Formula:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Meaning:

- Square root of MSE → brings the error back to the **same units as the target variable**.
- Easy to interpret compared to MSE.

Example:

- Random Forest RMSE = 12.87 → typical error in same unit as your BP/PPG readings.

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4. R^2 – Coefficient of Determination

Formula:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

- $\bar{y}$ = mean of actual values

Meaning:

- Shows **how much of the variance** in your target variable is explained by the model.
- Ranges from -∞ to 1.
  - 1 → perfect prediction
  - 0 → model predicts just as well as the mean
  - <0 → worse than predicting the mean

Example:

- Random Forest R^2 = 0.2416 → model explains **24% of the variance**.
- BiLSTM R^2 = 0.1078 → explains only 10% → weak performance.

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5. MAPE – Mean Absolute Percentage Error

Formula:

$$\text{MAPE} = \frac{100}{n} \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Meaning:

- Gives error as a **percentage of the actual value** → easy to understand relatively.
- Lower MAPE → better.

Example:

- Random Forest MAPE = 8.83% → on average, predictions are **8.83% off from true value**.

How to Compare Models Using These Metrics

1. **Look at errors (MAE, RMSE, MSE, MAPE)**
   - Smaller values = better predictive accuracy.
2. **Look at R^2**
   - Higher = model explains more variance in your data.
3. **Consistency**
   - A good model usually has **low errors AND higher R^2**.
   - Sometimes a model has low MAE but low R^2 → predicts close to mean, but cannot capture trends.

Quick Interpretation of Your Table

Metric	Best Model	Why
MAE	Random Forest	Lowest average error
MSE/RMSE	Random Forest	Few large errors, good overall fit
R^2	Random Forest	Explains most variance
MAPE	Random Forest	Most accurate relatively

✅ So **Random Forest clearly performs best** overall.

If you want, I can also **draw a simple diagram showing how these metrics relate to predictions vs actual values**, so it becomes really visual and easy to remember.

Do you want me to do that?

please explain these based on my reseach

Complex Problem Solving

EP1 Depth of Knowledge: The project needed to be aware of multiple fields such as electronics, biomedical signal processing, artificial intelligence, and software engineering knowledge. To construct the IoT device, it was necessary to learn physics of optical sensors and program microcontrollers, and preprocessing the signals of PPG required mathematical and statistical knowledge. To select, train, and optimize the models, machine learning knowledge was required. All these areas indicate that the problem could not be resolved in a single sphere of expertise, so EP1 requirements would have been fulfilled.

EP2 Range of Conflicting requirements: This project had various contradictory requirements. Our system had to trade-off between accuracy and real-time execution, as the very models like BiLSTM which were more predictive-accurate, were also computationally expensive. In a similar manner, data privacy and security would have to be provided without creating undue latency in data flow. Simplicity of user interface was a user requirement that was in conflict with having multiple layers of technicality. Handling these trade offs demanded an iterative approach to decision-making which embodies the spirit of EP2.

EP3 Depth of analysis : The project consisted of detailed analysis at several levels in order to create a functional BP prediction system. The selected waveforms of raw PPG were thoroughly studied by the time and frequency concepts by peak detection, skewness, kurtosis, and Fourier transformation. Statistical trends were analyzed to know their applicability to blood pressure. In addition to signal analysis, machine learning models were compared by quantitative assessment measures (RMSE and R2). This level of analysis outlines the multi-level analysis of the project as a subset of EP3.
EP4: Even though the application of PPG to predict cuffless blood pressure is a challenge issue reported in the biomedical literature, PPG is a clinical issue yet to be solved. Old problems like sensor noise, subject-to-subject variation and demographic variations in vascular response had to be considered. Meanwhile, a series of new problems was identified within the context of the real-time implementation of IoT hardware combined with web systems. Therefore, the project had to deal with known and unfamiliar issues, which is what meets the definition of EP4.

EP5 Extent of Applicable Codes: Even though the project was not designed as a certified medical device, it had to follow the associated standards and codes. This involved the legislation on personal health information protection, IoT security standards in transmitting via a Wi-Fi network, and biomedical signal processing standards regarding the processing of PPG data. Moreover, it was required to follow common machine learning guidelines like the separation of databases, assessment of the models, and reproducibility. These considerations demonstrate to what extent applicable codes extend to place the project in EP5.

EP7 Interdependence: The solution had a high relational aspect with components. The accuracy of features requires reliable PPG data acquisition of the driving element, which is the IoT device. These would have to be properly prestructured to feed machine learning models, the effectiveness of which directly influenced the user interface experience. Even a failure of any of the layers (either hardware, software, or AI) would lead to a failure of the overall system as well. Such interdependence of different subsystems is a good example of EP7 hence the problem is interdependent and complex at the same time.

Mapping with Knowledge Profile

K2 Mathematics: This project heavily relied on mathematics that supported the preprocessing, as well as modeling. Calculation of Fourier transforms, skewness and kurtosis and variance were all based on mathematical formulations. The process of training machine learning models was a regression-based optimization problem, and their assessment (MAE and RMSE) was founded on mathematical formulas to assess the importance of error. Signal analysis, as well as performance evaluation of a model, could not have occurred without this mathematical background. Therefore, mathematics (K2) was inescapable as far as project results are concerned.

K3 Engineering Fundamentals: Behind the design of the IoT-based devices and the communication system, there were the core engineering principles. Electrical circuits had to be understood, to connect the sensor, MAX30102 sensor, to ESP32 microcontroller. The concept of realizing embedded systems enabled real time signal capture, whereas the field of networking enabled the transfer of data wirelessly. Engineering principles were used to make sure the hardware worked in harmony with the backend and the web systems, which was the backbone of the whole solution. Therefore, the project very much coincides with K3.
K4: A key attribute to the success of the project was specialist knowledge. In order to create meaningful information out of noise in PPG waveforms biomedical signal processing knowledge was needed. High-tech AI concepts and ML concepts such as Random Forests and BiLSTM networks required expert computational expertise to run and optimize. Also, to provide real-time interaction between software and hardware, IoT systems expertise was required. This dependency on expert knowledge emphasises that the mapping of the project to K4 occurs.

K5 Engineering desgin: The project represented an example of an engineering design because it incorporated several subsystems into a solution. They needed more than just workable parts, but the system needed to be usable, efficient, and real time performance. This involved developing the casing of the IoT, organising the MongoDB structure and a visual data web interface. All design decisions weighed user requirements against technical feasibility, which is a characteristic feature of engineering design and mapping that is easily mapped to K5.

K6 Engineering Practices: In the invention of the iterative implementation and testing stage, the engineering practice was displayed. The hardware prototypes of the IoT device were refined a couple of times. Multiple versions of the data set were tested on preprocessing pipelines to test its robustness and AI models were trained, evaluated, and compared. The practical aspect of engineering work was embodied by debugging a bug in a code or a bug in a hardware. This embodiment of the theoretical design in practical working mechanisms is an expression of the K6 requirements.

K7 Comprehension: This project was something that had to be understood and analyzed at many levels of knowledge. Surveying literature about cuffless BP estimation aided expert positioning of the issue, whereas processing datasets and successfully performing feature extractions made it possible to achieve insight. There was also the need to understand how to integrate biomedical knowledge with AI and IoT. The project could not have been developed without this ability to understand and critically examine cross domains. This puts a strong sense of alignment with K7.

K8 Research Literature: The type of research literature available was so valuable in the methodology and system design. The features and models were selected based on studies on cuffless BP estimation using PPG.[6][9][15]. IoT healthcare reviews also had an impact on the system architecture. Using peer-reviewed literature allowed making certain that the project was not another quiet study, but it can be traced back to and justified by previous studies, which, in turn, pointed to K8.

Mapping with Complex Engineering Activities

EA1 Range of Recourses : A large variety of resources were used in this project, physical (ESP32, MAX30102, LCD, Omron sphygmomanometer) as well as software (Python, TensorFlow, Node.js, MongoDB). Further, Google Colab was an environment with cloud-based model training. These mixed resources demanded technical flexibility, and close coordination to implement. The depth of resources is a measure of complexity in a project and illustrates EA1.

EA2 Level of Interaction: In this project, there was interaction on various levels. IoT interaction with the web interface was facilitated by backend APIs that connected the IoT device to the system-to-system interaction. The frontend interface was the method to provide human-computer interaction, enable entry of demographical data, and provide real-time visualization of predictions. Within the ML models, interactions were also possible as features interacted with each other to make predictions. The multidimensional interrelationship between and among systems depicts EA2.

EA3 Innovation: The project innovation can be described as a combination of IoT hardware, signal processing using PPG, and machine learning in the form of a single real-time foretelling system. Although some of these individual components have been explored in the past, their integration to create a deployable and easy-to-use system that measures cuffless BP is new. In this strategy, we can see how engineering innovation can be achieved through new theories as well as through integrating the existing technologies into viable solutions thus fitting within EA3.

EA4 Consequences of for society and enverment: The system has a high degree of implications on the society. It is a low-cost, portability-based solution to the early detection and management of hypertension, one of the largest health-related problems worldwide. It can help patients gain active control of their health, which may help minimise healthcare visits and chronic problems, as it allows real-time tracking and monitoring. The IoT device has a minimally high carbon footprint, and it can become broadly available in the future due to scalability. Such extended implications are an example of EA4.

EA5 Familiarity: The project is neither too familiar nor too novel. Common components were generic IoT device development, signal preprocessing, and well-known machine learning procedures. Adoption of the These elements made a novelty by integrating into a single BP monitoring pipeline. The balance of it rendered the problem familiar enough and yet innovative. It therefore projects well to EA5.

Got it! Let's **map your research project metrics and model evaluation into the engineering problem-solving (EP), knowledge (K), and engineering activity (EA) framework** in a way that connects directly to your work with PPG-based cuffless blood pressure prediction. I'll explain each metric (MAE, RMSE, R^2, MAPE) in the context of your project and tie it to the EP/K/EA points you wrote.

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1. MAE – Mean Absolute Error

Context in your research:

- MAE tells you **on average how far your predicted BP is from the actual value**.
- Random Forest MAE = 9.08 → this means that on average, the predicted BP is **±9 mmHg off**, which is quite reasonable for a non-invasive IoT-based system.

Connection to EP/K/EA:

- **EP1 (Complex Problem Solving):** Choosing MAE as a metric required understanding **statistics and AI model evaluation**, showing cross-disciplinary knowledge (math + AI).
- **K2 (Mathematics):** Calculation of MAE directly involves basic statistical operations.
- **EA2 (Level of Interaction):** The error metric shows **how feature interactions and model choices impact overall accuracy**.

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2. RMSE – Root Mean Squared Error

Context:

- RMSE penalizes **large errors more than MAE**.
- Your Random Forest RMSE = 12.87 → indicates that occasionally, the system could make larger mistakes (~13 mmHg) even if average MAE is smaller.

Connection to EP/K/EA:

- **EP2 (Range of Conflicting Requirements):** RMSE highlights trade-offs: some models like BiLSTM have better theoretical accuracy but are computationally expensive → showing conflict between **accuracy vs real-time performance**.
- **K2 (Mathematics):** Involves squaring errors, summing, and square rooting → shows applied math in performance evaluation.

- **EA1 (Range of Resources):** RMSE helps assess how different resources (data preprocessing, IoT sensor precision) affect prediction accuracy.

---

3. R^2 – Coefficient of Determination

Context:

- R^2 measures **how much of the variance in blood pressure your model explains**.
- Random Forest $R^2 = 0.2416$ → the model explains ~24% of variability; low R^2 shows the **challenge of predicting BP from PPG**, considering biological variation, sensor noise, and demographic differences.

Connection to EP/K/EA:

- **EP4 (Known and Unfamiliar Problems):** Low R^2 reflects **subject-to-subject variability** and the real-world complexity of non-invasive BP estimation.
- **K7 (Comprehension):** Understanding why R^2 is low requires **critical analysis of signals, features, and model limitations**.
- **EA5 (Familiarity):** The low R^2 shows partial familiarity—the problem uses standard techniques, but real-world variability makes it partially unfamiliar.

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4. MAPE – Mean Absolute Percentage Error

Context:

- Expresses **error as a percentage of actual BP**, making it easier to understand relative performance.
- Random Forest MAPE = 8.83% → good for a practical system that non-experts may use.

Connection to EP/K/EA:

- **EP3 (Depth of Analysis):** Using MAPE along with MAE and RMSE shows **multi-level performance evaluation**: absolute, squared, and relative error.
- **K2/K4 (Mathematics + Specialist Knowledge):** MAPE requires understanding ratios and percentage errors, which are applied in evaluating biomedical signals.
- **EA4 (Consequences for Society):** A low MAPE ensures predictions are relatively accurate for patients, which has **direct societal impact**.

Bringing it all together

Your **Random Forest model** is the best because:

1. **MAE & MAPE are lowest** → practical, real-world performance is good.
2. **RMSE is low** → large errors are controlled.

3. **R^2 is highest** → explains more variance than others, even if still limited due to biological variability.

Engineering and research significance:

- Choosing the right metrics (MAE, RMSE, R^2, MAPE) required **interdisciplinary knowledge (EP1, K2, K4)**.
- Balancing **accuracy, real-time execution, and computational cost** reflects **conflicting requirements (EP2, EA2)**.
- Evaluating multiple models using these metrics reflects **depth of analysis (EP3, K6)**.
- Understanding the societal implications of your error metrics maps to **EA4**.

If you want, I can **make a clean table or visual that links each metric to EP, K, and EA directly** for your report—so your evaluators can see it clearly.

Do you want me to do that?

how do this relate to my reseach based project

hey, i am working on a Research-based IoT-web project.
i have worked a lot in this project,
things done,
Frontend Done
Backend Code done
IoT code done
dataflow confirmed
data collection
data pre processing
AI model testing
casing created
final deployable model creatation
the documentation/ reseach report
making website depoloment of the model
website intigration

the idea,
using a IoT device: sensor MAX30102, microcontroller esp32, and a small lcd(2x16), small button to start the process. it takes data reading for about 30 seconds, then finds the Heart rata, SPO2, IR values in a array, RED values in a array, and send that data to a MongoDB database.

then in the website, we fetch the latest data, from the IoT device, add demographic data, such as, age, gender, height, weight and BMI(derived). and for the data collection part, the blood pressure values: systolic and diastolic values as the decision class and update the MongoDB database.

we collect data like that,

after we collect enough data, we will download a CSV file, from mongoDB or make a CSV file from the json file.

then, preprocess the data, so that, we can easily build a AI model that can output, Blood Pressure values from the sensor data an Demographic data.

for the preporcessing part, i from about 168 IR array value and another 168 RED array value, i feture extracted 15 fetues such as skewness, kurtius and pean_number and other fetures.

later adter analysising the data via model training i realzed red value and ir value trining at the same time is redundent and it was lowering the resutls overall, so i only used the IR value fetures to train the models.

from that i found out that my data colletion is flawed, due to sensor quality and data bais on aga and gender.

too many data on the 20-25 years age almost 50% and almost 70% male data.

due the data best R^2 i got fomr the regression models was 24%, and MAE of 9.2 and that was Random forest.

dispite that, i deoplyed that model using flask api and hungging face as the host.

then deploy the AI model in the same Website, different page.

the frontend is done using HTML and Javascript

the backedn server code is done using Node.js and MongoDB as database, while using Vercel as the hosting dommain.

also doing lit review and similar works, and gap analysis. for this.

Perfect! Now I can clearly connect your **research-based IoT-web project** to the **metrics (MAE, RMSE, R^2, MAPE), EP (Engineering Problems), K (Knowledge), and EA (Engineering Activities) framework**. Let's break it down so it's obvious how everything relates to your work.

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1. How your model evaluation metrics relate to your project

Metric	What it shows	How it relates to your project
MAE (Mean Absolute Error)	Average difference between predicted BP and actual BP	Shows how close your deployed Random Forest predictions are to real-world BP readings from your IoT device. Lower MAE (9.2) indicates reasonably accurate predictions despite noisy sensor data and demographic bias.
RMSE (Root Mean Squared Error)	Penalizes larger prediction errors	Helps you understand occasional big mistakes in BP predictions. Critical because large errors could misinform users about hypertension risk.

Metric	What it shows	How it relates to your project
R^2 (Coefficient of Determination)	How much variance in BP is explained by your model	Reflects the limits of predicting BP from PPG signals and demographic features. Your R^2 = 0.24 highlights data quality issues (sensor noise, biased sampling) and shows the **challenge of cuffless BP prediction**.
MAPE (Mean Absolute Percentage Error)	Average percentage error	Shows practical, real-world relative accuracy. For a user, an 8–9% error rate is interpretable: the system is not perfect, but usable for non-clinical monitoring.

✅ In short, these metrics **quantify the performance and limitations of your deployed AI model**, which is central to your research-based project.

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2. How this connects to EP (Engineering Problems)

EP	Connection to your project
EP1 (Complex Problem Solving)	Your project integrates electronics (MAX30102, ESP32), signal processing (IR/RED array, feature extraction), AI/ML models, and software engineering (Node.js, HTML/JS frontend, Flask API). Multiple domains were required, fulfilling EP1.
EP2 (Range of Conflicting Requirements)	Trade-offs: accuracy vs real-time processing, using BiLSTM vs Random Forest; privacy/security vs low latency; simple UI vs technical depth. Your project handled these conflicts iteratively.
EP3 (Depth of Analysis)	Detailed signal preprocessing, feature extraction (skewness, kurtosis, peak number), and model comparison via RMSE, R^2, MAE. You analyzed features and models deeply.
EP4 (Known and Unfamiliar Problems)	Challenges like sensor noise, demographic bias, and real-time IoT deployment represent **both known (literature) and new issues**.
EP5 (Extent of Applicable Codes)	Followed IoT security standards, data privacy considerations, and machine learning reproducibility practices.
EP7 (Interdependence)	Accuracy depends on hardware (sensor), software (preprocessing pipeline), AI model, and frontend integration. Any failure affects the whole system.

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3. How this connects to K (Knowledge Profile)

K	Connection
K2 (Mathematics)	Feature extraction (skewness, kurtosis, Fourier transforms), model evaluation (MAE, RMSE, R^2, MAPE), and regression training all require math/statistics.
K3 (Engineering Fundamentals)	IoT hardware design (ESP32, MAX30102, LCD), circuits, networking, and backend/frontend integration.
K4 (Specialist Knowledge)	AI/ML models (Random Forest, BiLSTM), PPG biomedical signal processing, real-time IoT data handling.
K5 (Engineering Design)	Created a **complete system**: IoT sensor casing, database structure, web interface, deployable AI model.
K6 (Engineering Practices)	Iterative testing: refining prototypes, preprocessing pipelines, AI model tuning, debugging hardware/software.
K7 (Comprehension)	Understanding multiple domains (AI, biomedical, IoT) and integrating them effectively.
K8 (Research Literature)	Reviewed literature on cuffless BP estimation, IoT healthcare, and ML-based BP prediction for methodology and gap analysis.

4. How this connects to EA (Engineering Activities)

EA	Connection
EA1 (Range of Resources)	Used physical resources (ESP32, MAX30102, LCD) and software resources (Python, TensorFlow, Node.js, MongoDB, Colab, Vercel).
EA2 (Level of Interaction)	Hardware → backend API → database → frontend; feature interactions in ML models.
EA3 (Innovation)	Integrated IoT, PPG signal processing, and ML into a **deployable real-time BP monitoring system**.
EA4 (Consequences for Society & Environment)	Low-cost, real-time BP monitoring can aid early detection of hypertension, reduce healthcare visits. IoT device has a minimal carbon footprint and can scale.
EA5 (Familiarity)	Used familiar components (IoT devices, ML models) but combined them innovatively into a single deployable system.

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5. Linking Metrics to EP/K/EA

- **MAE & MAPE → EP3, K2, EA4:** Show practical usability and societal relevance.
- **RMSE → EP2, K2, EA1:** Helps manage trade-offs and system resource implications.
- **R^2 → EP4, K7, EA5:** Reflects challenges of real-world, partially unfamiliar problems.

In other words, **your metrics aren't just numbers—they demonstrate your project's complexity, the interdisciplinary knowledge applied, and the impact of your solution.**

If you want, I can **draw a diagram showing your whole IoT-ML project workflow with where each metric, EP, K, and EA fits**—it'll make your defense or report super clear.

Do you want me to make that?